

Homework #3

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2. For the remaining exercises in this set, we will use one of R's built-in data sets, called the "ChickWeight" data set. According to the documentation for R, the ChickWeight data set contains information on the weight of chicks in grams up to 21 days after hatching. Use the `summary(ChickWeight)` command to reveal basic information about the ChickWeight data set. You will find that ChickWeight contains four different variables. Name the four variables. Use the `dim(ChickWeight)` command to show the dimensions of the ChickWeight data set. The second number in the output, 4, is the number of columns in the data set, in other words the number of variables. What is the first number? Report it and describe briefly what you think it signifies.

```
c.weight<-ChickWeight
summary(c.weight)
```

```
##      weight      Time      Chick      Diet
##  Min.   : 35.0   Min.   : 0.00   13      : 12   1:220
##  1st Qu.: 63.0   1st Qu.: 4.00    9       : 12   2:120
##  Median :103.0   Median :10.00   20       : 12   3:120
##  Mean   :121.8   Mean   :10.72   10       : 12   4:118
##  3rd Qu.:163.8   3rd Qu.:16.00   17       : 12
##  Max.   :373.0   Max.   :21.00   19       : 12
##                                     (Other):506
```

```
dim(c.weight)
```

```
## [1] 578  4
```

```
head(c.weight)
```

```
##  weight Time Chick Diet
## 1     42    0     1     1
## 2     51    2     1     1
## 3     59    4     1     1
## 4     64    6     1     1
## 5     76    8     1     1
## 6     93   10     1     1
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.3      v readr      2.1.4
```

```
## v forcats 1.0.0      v stringr 1.5.0
## v ggplot2 3.4.4      v tibble 3.2.1
## v lubridate 1.9.3    v tidyr 1.3.0
## v purrr 1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag() masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
glimpse(c.weight)
```

```
## Rows: 578
## Columns: 4
## $ weight <dbl> 42, 51, 59, 64, 76, 93, 106, 125, 149, 171, 199, 205, 40, 49, 5~
## $ Time <dbl> 0, 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 21, 0, 2, 4, 6, 8, 10, 1~
## $ Chick <ord> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 2, 2, 2, 2, 2, 2, 2, ~
## $ Diet <fct> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ~
```

The four variables in the data set are: weight, Time, Chick, and Diet. The dimensions of the data set are 578x4, where 4 represents the number of columns and 578 represents the number of rows, in this case each row representing the measurements of one chick.

3. When a data set contains more than one variable, R offers another subsetting operator, `$`, to access each variable individually. For the exercises below, we are interested only in the contents of one of the variables in the data set, called `weight`. We can access the `weight` variable by itself, using the `$`, with this expression: `ChickWeight$weight`. Run the following commands, say what the command does, report the output, and briefly explain each piece of output: `summary(ChickWeight$weight)` *edited question for use in markdown

```
summary(c.weight$weight)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      35.0   63.0   103.0   121.8   163.8   373.0
```

```
# Summary() gives us the statistical five-number summary for the weight column specified by c.weight$weight, so
head(c.weight$weight)
```

```
## [1] 42 51 59 64 76 93
```

```
# When using head() it displays the first few entries of whatever is selected. In this case since we have
mean(c.weight$weight)
```

```
## [1] 121.8183
```

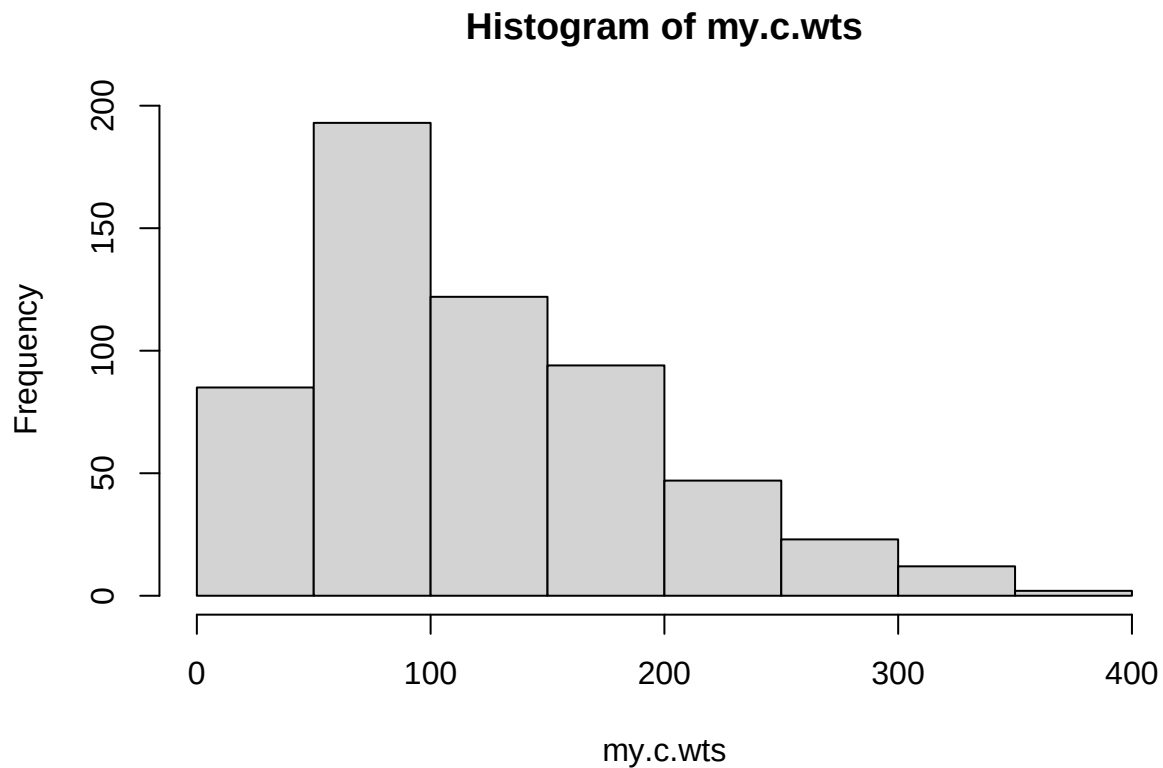
```
# mean() displays the mean of whatever value is inside of (), in this case we are displaying the mean of
my.c.wts<-c.weight$weight
# The above command generates a new variable called my.c.wts which is made of the c.weight$weight column
quantile(my.c.wts,0.50)
```

```
## 50%
## 103
```

```
# Quantile shows the specified quantile, here we specified the 0.50 quantile which is the median and gives
```

4. In the second to last command of the previous exercise, you created a copy of the weight data from the ChickWeight data set and put it in a new vector called myChkWts. You can continue to use this myChkWts variable for the rest of the exercises below. Create a histogram for that variable. Then write code that will display the 2.5% and 97.5% quantiles of the distribution for that variable. Write an interpretation of the variable, including descriptions of the mean, median, shape of the distribution, and the 2.5% and 97.5% quantiles. Make sure to clearly describe what the 2.5% and 97.5% quantiles signify.

```
hist(my.c.wts)
```



```
quantile(my.c.wts, c(0.025,0.975))
```

```
##      2.5%   97.5%  
##  41.000 294.575
```

```
mean(my.c.wts)
```

```
## [1] 121.8183
```

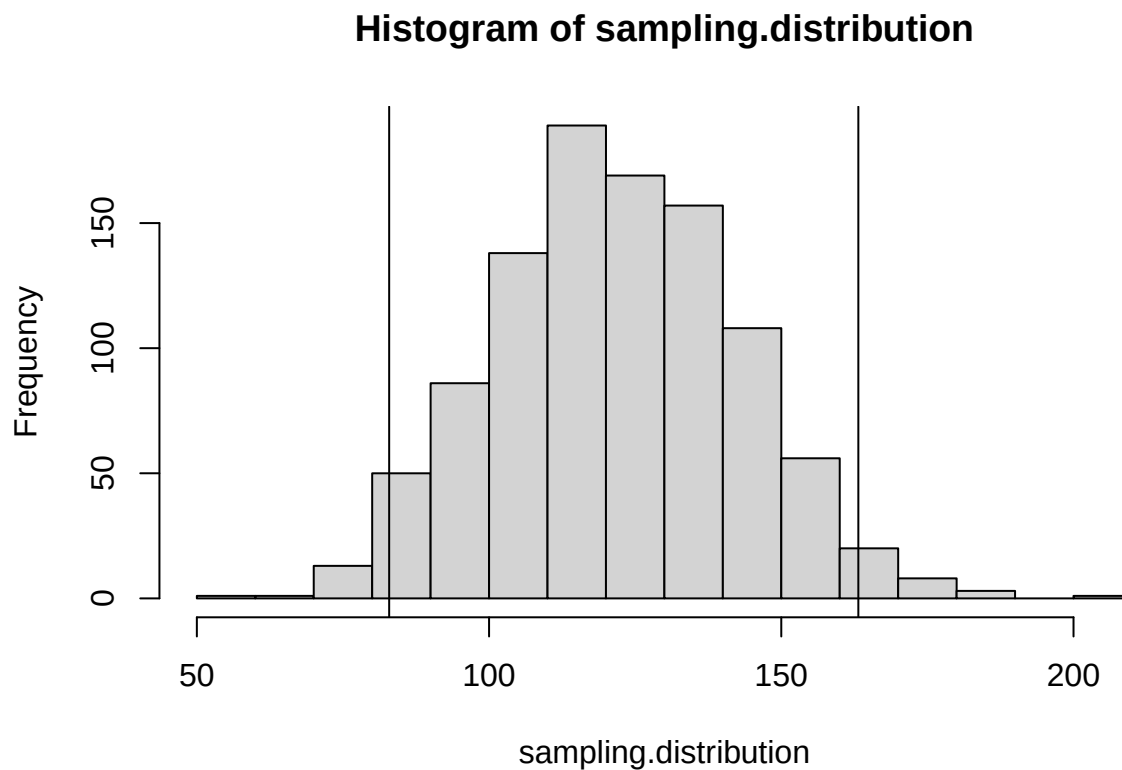
```
median(my.c.wts)
```

```
## [1] 103
```

From viewing the histogram of `my.c.wts` we can see that there is a rightward skew to the histogram which most likely means that the mean is to the right of the median which is caused by the skew in the histogram. We can verify this by looking at the values for the mean and median for `my.c.wts`, where the mean is ~ 121 and the median is ~ 103 , which confirms the mean is to the right of the median. When looking at the output for the `quantile()` we can see the lower 2.5% and upper 97.5% of the values. These values represent the tails of the data and combined account for the outer 5% of values, where the 67-95-99 rule comes from, the 2.5% and 97.5% values represent everything beyond the second quartile, or in other words, the tails of the data.

5. Write R code that will construct a sampling distribution of means from the weight data (as noted above, if you did exercise 3 you can use `myChkWts` instead of `ChickWeight$weight` to save yourself some typing). Make sure that the sampling distribution contains at least 1,000 means. Store the sampling distribution in a new variable that you can keep using. Use a sample size of $n = 11$ (sampling with replacement). Show a histogram of this distribution of sample means. Then, write and run R commands that will display the 2.5% and 97.5% quantiles of the sampling distribution on the histogram with a vertical line.

```
sampling.distribution<-replicate(1000,mean(sample(my.c.wts, size=11, replace=TRUE)), simplify=TRUE)
hist(sampling.distribution)
abline(v=quantile(sampling.distribution,0.025))
abline(v=quantile(sampling.distribution,0.975))
```



Performed instructions as listed in question.